Political Classification of OCRed Campaign Momentos

ARChELeC corpus – French legislative elections of 1993

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Abstract

This report studies whether the left–right political orientation of candidates can be recovered from OCRed electoral momentos from the ARChELeC corpus, focusing on the first round of the 1993 French legislative elections. Since the original OCRed texts do not include ideological labels, I construct a supervised binary dataset from structured political metadata, mainly the candidate's political support and electoral list, with a limited cross-check using ParlGov. After filtering, political mapping and cleaning, the final corpus contains 2,523 momentos labelled as *left* or *right*. I compare classical text classification models using Bag-of-Words and TF–IDF representations, run a de-biasing experiment to remove explicit party and ideological cues, and test recurrent neural models (LSTM and GRU) on the de-biased corpus. The strongest baseline is a TF–IDF + linear SVM model, which reaches an accuracy and macro-F1 of 0.99 on the held-out test set. The de-biasing experiment shows that performance does not rely only on party names or direct ideological markers. Recurrent models capture a political signal but remain below the linear baseline. Finally, a named-entity recognition analysis complements the classification results by identifying the actors, organisations and places most frequently mentioned by each political orientation. Overall, the results show that the left–right divide is strongly reflected in the lexical and referential structure of these historical electoral momentos.

1 Introduction

In many democracies, political competition is commonly organised around a left–right divide. This distinction helps voters interpret parties, candidates and electoral choices, even when campaigns also involve local issues, personal trajectories and specific policy proposals. In this context, electoral momentos constitute a valuable source for studying how candidates present themselves and how political orientations are expressed in campaign discourse.

In the French case, the ARChELeC corpus provides access to historical electoral documents from the Fifth Republic, including OCRed momentos from the 1993 legislative elections Sciences Po Library [2023], Gaultier-Voituriez [2022]. These documents are particularly informative because they combine programmatic arguments, party affiliations, ideological references and local campaign themes. They therefore make it possible to study political discourse at the candidate level, beyond national party platforms or parliamentary debates. More broadly, this work belongs to the growing literature that uses political texts as data to analyse discourse and political competition at scale Grimmer and Stewart [2013].

This report asks whether the left–right divide can be automatically recovered from electoral momentos. The objective is to predict, from the text of a candidate's momento, whether the candidate belongs to the left or the right political space. To address this question, the project builds a supervised NLP pipeline using labels reconstructed from political metadata, especially the candidate's

political support and electoral list. The analysis then compares classical text classification models, a de-biasing experiment and a named-entity recognition approach. The goal is not only to obtain accurate predictions, but also to examine whether political orientation is reflected in meaningful lexical and discursive patterns.

2 Corpus and target construction

2.1 Data sources and selected subset

This project relies on two complementary sources of data: the OCRed texts of electoral momentos and the structured metadata provided with the ARCHELEC corpus. The OCR files constitute the textual material used by the NLP models. Each file corresponds to the text automatically extracted from a scanned electoral document. These momentos contain the name of the candidate, political affiliations, political proposals, slogans, calls to vote, and lists of policy commitments. They are therefore well suited to studying political discourse at the candidate level.

The second source is the ARCHELEC metadata table. It contains, for each OCR file, the document identifier, the year of the election, the type of election, the round, the department and constituency, as well as political information associated with the candidate. Two fields are especially important for this project: *titulaire-soutien* and *titulaire-liste*. The first indicates the political support or movement associated with the candidate, while the second indicates the electoral list or coalition under which the candidate ran.

The analysis focuses on **the first round of the 1993 French legislative elections**. This choice is motivated by the size and coherence of the available data. The 1993 subset contains many OCRed documents, which makes it possible to build a sufficiently large supervised dataset. At the same time, restricting the study to a single election year avoids mixing different electoral contexts, party configurations and campaign vocabularies. The task is therefore defined within a coherent political moment.

2.2 Selected subset and construction of the target variable

The initial OCR archive for the 1993 legislative election contained **5,840 files**. I focus on the first round of this election because it provides a large and politically coherent subset. After filtering the metadata to keep only the 1993 legislative election, first round, **4,872 rows** were retained and matched with the OCR files using the document identifier.

The objective of the project is to predict whether a candidate belongs to the *left* or the *right* political space. Since the raw OCRed textos do not contain a ready-to-use ideological label, a supervised target variable had to be constructed from external information.

Metadata used for labelling. The target variable was constructed mainly from two metadata fields: *titulaire-soutien* and *titulaire-liste*. This choice is important because the OCRed text remains the predictive input, whereas the label is primarily derived from external metadata rather than directly from the text.

The labelling procedure followed three steps:

1. extraction of all distinct combinations of political support and electoral list;
2. mapping of these combinations into three categories: *left*, *right*, or *other*;
3. exclusion of ambiguous or unreliable observations from the final binary dataset.

Political mapping. Clear affiliations with left-wing parties or movements were assigned to the *left*, for example *Parti socialiste*, *Parti communiste français*, *Lutte ouvrière* or *Ligue communiste révolutionnaire*. Conversely, affiliations with the parliamentary right, centre-right, Gaullist parties or the far right were assigned to the *right*, for example *Rassemblement pour la République*, *RPR*, *Union pour la démocratie française*, *UDF*, *Union pour la France*, *Front national* or *divers droite*.

Ambiguous, independent, non-aligned, regionalist, ecological or insufficiently informative cases were assigned to *other*. This includes candidates labelled as *sans étiquette*, lists explicitly described as neither left nor right, and ecological coalitions with mixed or unclear positioning.

Additional correction rules were applied to handle specific cases. Formations clearly associated with the moderate or centre-right, such as *Centre des démocrates sociaux*, *Démocratie chrétienne française*, *Parti républicain*, or labels containing *centre droit*, were assigned to the *right*. Some cases with missing or ambiguous political support were assigned to the *left* when the electoral list explicitly referred to the left, for example through expressions such as *à gauche*, *changer à gauche*, or *oxygéner la gauche*. Finally, lists labelled as *opposition* were assigned to the *right* when they were combined with right-wing supports such as *UDF*, *RPR*, *gaulliste*, or *divers droite*.

Reliability checks. After this metadata-based mapping, only observations labelled as *left* or *right* and marked as reliable were retained. In a final cleaning step, a limited cross-check with ParlGov[1] was used to correct or confirm some party-based labels.

This step did not create the labels from scratch. It used the already labelled binary dataset and searched the OCRed text for a small set of explicit party mentions, such as *Parti socialiste*, *RPR*, *UDF* or *Front national*. When such a party was detected, its orientation was compared with the external ParlGov classification and the final label was updated if necessary. The final dataset therefore relies mainly on metadata-derived labels, with a limited text-based correction using ParlGov.

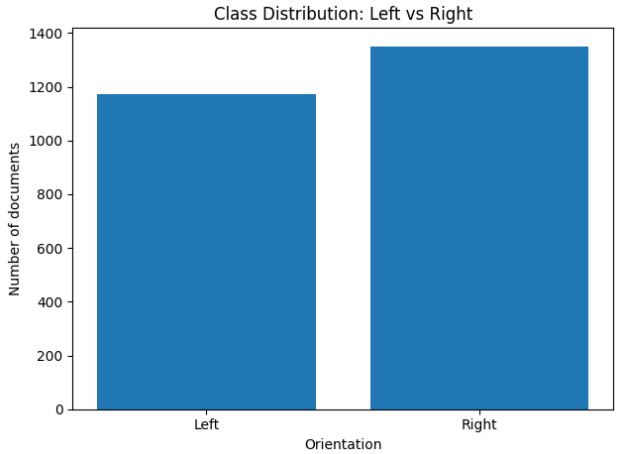
Final binary corpus. After political mapping and cleaning, the final binary dataset contains **2,523 documents**: **1,172** labelled as *left* and **1,351** labelled as *right*. The distribution is slightly imbalanced in favour of the right, but the imbalance remains moderate.

Table 1: Construction of the working corpus.

Step	Description	Volume
OCR archive	OCR files for the 1993 legislative election	5,840 files
Metadata filtering	1993 legislative election, first round	4,872 rows
OCR–metadata matching	Join by document identifier	4,872 files
Political mapping	Assignment to left, right or other	4,872 files
Binary subset	Kept left/right cases only	2,653 files
Final dataset	Removal of duplicates and unstable cases	2,523 files

Note: This table summarises the main steps used to transform the initial OCR archive and metadata into the final binary corpus used for classification.

Figure 1: Distribution of documents by political orientation after labelling.



Note: This figure shows the class distribution in the final binary dataset, with 1,172 momentos labelled as *left* and 1,351 momentos labelled as *right*.

[1]ParlGov is an external database on parties, elections and cabinets in established democracies. See `https://www.parlgov.org/`.

2.3 Discussion of the limitation of the labellisation

An inherent risk of this labelling process described in Section 2.2 is that the texts themselves contain explicit political cues (party names, mentions of "left" or "right"). The model could therefore learn to recognise these markers without capturing deeper discursive differences. To limit this information leakage I analyse the most influential words and run a de-biasing experiment in which explicit markers are removed (see Section 3.3).

3 Methodology

This section describes the complete methodological pipeline used to transform the OCRed electoral momentos into a supervised text classification task. The pipeline includes text preprocessing, feature representation, classical classification models, a robustness experiment based on de-biasing, sequential neural models, and a complementary named-entity recognition analysis.

3.1 Text preprocessing and feature representation

The final dataset used in the experiments contains three main columns: the document identifier, the OCRed text of the momento, and the binary political label. Since the corpus comes from OCRed electoral documents, the raw texts contain several types of noise: missing accents (*démocratie* rendered as *democratie*), irregular spacing (*Union pour la France* rendered as *Union pour la France*), segmentation errors (*Rassemblement* split as *Rassem- blement*), spelling or typographical variants (*RPR* also appearing as *R.P.R.*), repeated documentary fragments (*élections législatives – 21 mars 1993* or *Sciences Po / Fonds CEVIPOF*), and duplicated or near-duplicated texts (the same flyer appearing twice with minor OCR or layout differences). These issues can increase vocabulary sparsity, make tokenisation less consistent, and create redundant or unstable observations.

During dataset construction, empty OCR files, duplicated documents and unstable[2] cases are removed. The preprocessing strategy is deliberately **light**. I apply a simple textual normalisation step in order to make the OCRed texts more consistent across documents. This normalisation consists of lower-casing the text, replacing newline characters with spaces, collapsing repeated spaces, and removing simple layout artefacts. However, I do **not** apply aggressive spelling correction, systematic accent normalisation or lemmatisation. This is because automatic correction may alter party acronyms, candidate names, local references or campaign slogans, which are precisely the kinds of lexical cues that can carry political information in electoral momentos.

Stopwords are also **not removed** in the main classical pipeline. This is a conservative choice. Generic stopword lists may remove words that contribute to campaign rhetoric, such as pronouns, negations or recurrent expressions. In addition, TF–IDF already down-weights very frequent terms, so keeping stopwords preserves stylistic information without giving excessive weight to the most common words.

For the classical models, I use two standard text representations:

- **Bag-of-Words (BoW)** represents each momento by the frequency of its terms. It captures how often each word appears in a document, without taking word order into account.
- **TF–IDF** uses the same lexical basis but gives more weight to terms that are frequent in a given document and less frequent in the corpus as a whole. This reduces the influence of very common words and highlights more distinctive lexical items.

These representations are well suited to linear models because they produce sparse high-dimensional vectors and make it possible to inspect which words contribute most strongly to the prediction. The dataset is split using an 80/20 stratified train–test split, that is, 80% of the documents are used for training and 20% (505 documents) are held out for final evaluation.

[2] A case is considered *unstable* when the same cleaned text is associated with conflicting labels, or when duplicated versions of a document could create leakage between the training and test sets.

3.2 Classification models and evaluation metrics

The main task is a supervised binary classification problem about predicting whether a momento belongs to the *left* or the *right* political space. I first rely on classical machine learning models because they are simple and strong baselines for text classification. Three classifiers are compared, each with both Bag-of-Words and TF–IDF features:

- **Naive Bayes**: a fast probabilistic baseline that often performs well on sparse text data. It assumes conditional independence between features, which is a strong assumption but can still be effective in text classification.
- **Logistic regression**: a linear classifier that is useful not only for prediction, but also for interpretation. Its coefficients make it possible to identify words associated with each political class.
- **Linear Support Vector Machine (SVM)**: a margin-based classifier particularly well suited to high-dimensional sparse representations. It is expected to perform well when the classes are separated by strong lexical patterns.

This comparison allows to identify the strongest predictive baseline for the corpus and provides interpretable evidence about the lexical structure of political discourse before moving to more complex models.

Model performance is evaluated using accuracy, precision, recall, F1-score and the confusion matrix. Accuracy is informative because the class imbalance is moderate, but it is not sufficient on its own. Precision, recall and F1-score are therefore reported for both classes in order to verify that high performance is not driven only by the majority class.

3.3 Robustness and de-biasing

A central methodological issue is the possible presence of explicit political cues in the momentos. The labels are constructed from political metadata, especially the candidate's political support and electoral list. Some of this information may also appear directly in the OCRed text. For example, a momento can mention party names such as *Parti socialiste*, *Parti communiste français*, *Front national*, *RPR* or *UDF*, as well as direct ideological terms such as *gauche* and *droite*. If the classifier relied only on these markers, the task would become partly trivial and the model would reveal little about broader political discourse.

To control for this risk, I conduct a **de-biasing experiment**. The procedure is as follows:

1. I first train the baseline linear models on the original corpus.
2. I inspect the most influential lexical features, especially the coefficients of logistic regression and the most discriminant SVM features.
3. I identify explicit political markers, including party names, party acronyms and direct ideological terms.
4. These terms are removed from the text to create a de-biased version of the corpus.
5. The same classification pipeline is retrained on this de-biased corpus.
6. Performance is compared with the original baseline.

This robustness check addresses the risk of **information leakage** between the metadata-derived labels and the textual input. If performance remains high after de-biasing, this suggests that the model captures broader lexical and discursive patterns, such as references to social issues, national identity, economic policy, work, solidarity, security or institutions, rather than only recognising party names.

3.4 Sequential models

In addition to classical models, I test two recurrent neural network architectures: an **LSTM**[3] and a **GRU**[4]. These models are implemented in PyTorch and trained on the de-biased version of the corpus. Their purpose is to evaluate whether models that take word order into account can improve over sparse lexical representations.

Before training, the de-biased texts are tokenised and converted into sequences of integer indices. A vocabulary is built from the most frequent tokens in the training set. Two special tokens are added: `<PAD>`, used to pad shorter sequences so that documents can be processed in batches, and `<UNK>`, used to represent words that are not included in the vocabulary, such as rare tokens or OCR-specific variants. Each recurrent model includes its own embedding layer, which converts token indices into dense vectors learned during training. These embeddings are then processed sequentially by either the LSTM or the GRU layer, and the resulting representation is passed to a final linear layer that predicts the label *left* or *right*.

Both models are trained with early stopping based on the F1-score measured on the validation set: training stops when this score no longer improves for a fixed number of epochs. If recurrent models do not outperform the classifiers, this suggests that the main predictive signal is already well captured by word frequencies and TF–IDF weights. Conversely, if they improved performance, this would indicate that sequential structure and word order contain additional information.

3.5 Named-entity recognition

Finally, I conduct a named-entity recognition analysis using a pre-trained French spaCy model. This step is not designed to improve the classifier directly. Instead, it provides an interpretive complement to the classification results.

The NER model extracts entities such as persons (PER), locations (LOC), organisations (ORG), miscellaneous named references (MISC). The objective is to examine whether left- and right-oriented texts differ not only in their vocabulary, but also in the actors, institutions and places they mention. For each document, the extracted entities are stored with their text span, entity type and political label. I then compute aggregate statistics by entity type and by political orientation.

This allows me to compare, for example, whether one side mentions more organisations, more places or more political figures. Since the NER model is not specifically trained on OCRed electoral momentos, its outputs must be interpreted with caution. OCR noise, historical names, party acronyms and layout artefacts can lead to incorrect entity boundaries or labels. For this reason, NER is used here as an **exploratory and interpretive tool**, rather than as a definitive measurement or as an input feature in the main classifier.

4 Results

4.1 Classical baseline models

The first set of experiments compares classical text-classification models using Bag-of-Words and TF–IDF representations. Overall, linear models perform extremely well on the corpus, while Naive Bayes is less stable, especially with Bag-of-Words features. Table 2 reports the main screening results.

The retained baseline is the **TF–IDF + linear SVM** model with an accuracy et a macro-F1 of 0.99. Although Logistic Regression with Bag-of-Words performs equally well and is also interpretable, the TF–IDF + SVM configuration is retained because TF–IDF down-weights very frequent terms and the linear SVM is particularly well suited to high-dimensional sparse text representations. The TF–IDF vectorizer is limited to **5,000 unigram features** and the linear SVM is trained with the default regularisation strength **C = 1.0**, a **linear decision function**, a maximum of **5,000 iterations**, and a fixed **random seed of 42**.

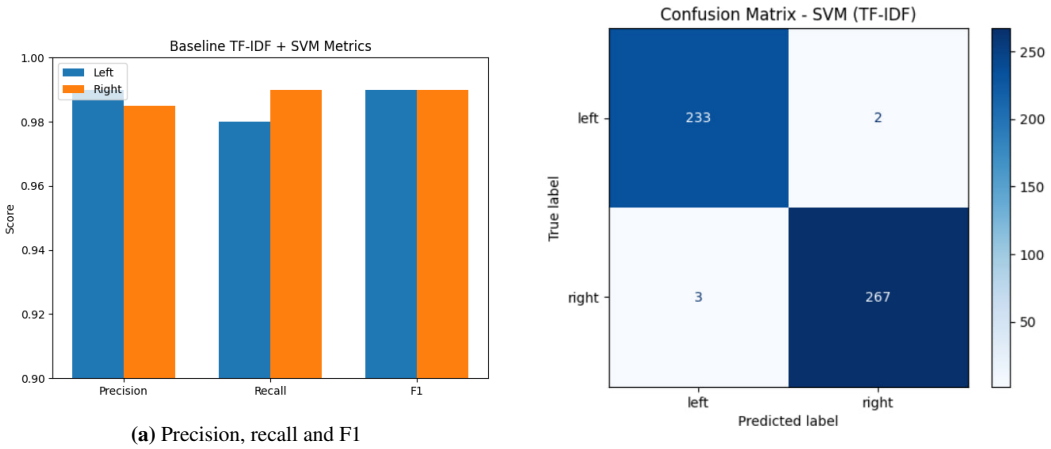
[3] Long Short-Term Memory: a recurrent neural network designed to process ordered sequences and capture long-range dependencies through memory gates.

[4] Gated Recurrent Unit: a recurrent neural network that also processes ordered sequences, but with a simpler gating mechanism and fewer parameters than an LSTM.

Table 2: Global classification metrics for classical baselines.

Rep.	Model	Accuracy	Macro Precision	Macro Recall	Macro F1
BoW	Log. Reg.	0.99	0.99	0.99	0.99
BoW	Naive Bayes	0.89	0.90	0.90	0.89
BoW	Linear SVM	0.99	0.99	0.99	0.99
TF–IDF	Log. Reg.	0.98	0.98	0.98	0.98
TF–IDF	Naive Bayes	0.95	0.95	0.95	0.95
TF–IDF	Linear SVM	0.99	0.99	0.99	0.99

The confusion matrix shows that it correctly classifies 233 left-wing momentos and 267 right-wing momentos. Only 2 left-wing documents are predicted as right-wing, and 3 right-wing documents are predicted as left-wing.



(a) Precision, recall and F1

(b) Confusion matrix

Figure 2: Performance of the TF–IDF + SVM baseline.

Note: Panel (a) reports precision, recall and F1 by class. Panel (b) shows the confusion matrix. Both confirm that the model performs almost perfectly on the held-out test set.

These results show that the left–right distinction is highly visible in the lexical structure of the corpus. The very small number of errors and their (almost) balanced distribution across classes also suggest that the model is not simply favouring the majority class.

4.2 Word analysis and de-biasing

The analysis of linear coefficients confirms that the model captures meaningful political vocabulary. The de-biasing experiment checks whether the model relies only on such obvious markers. To do so, the most explicit party names (e.g., *Parti socialiste*, *Parti communiste français*, *Front national*, *Rassemblement pour la République*, *Union pour la démocratie française*), party acronyms (e.g., *PS*, *PCF*, *FN*, *RPR*, *UDF*) and ideological terms (e.g., *gauche*, *droite*, *socialiste*, *communiste*) are removed from the corpus. The same TF–IDF + SVM pipeline is then trained again on the de-biased texts.

After this removal, **the macro-F1 remains equals to 0.99**. This means that the classifier does not rely only on direct party labels. Instead, it also captures broader lexical and discursive patterns in the way political issues, actors and priorities are formulated.

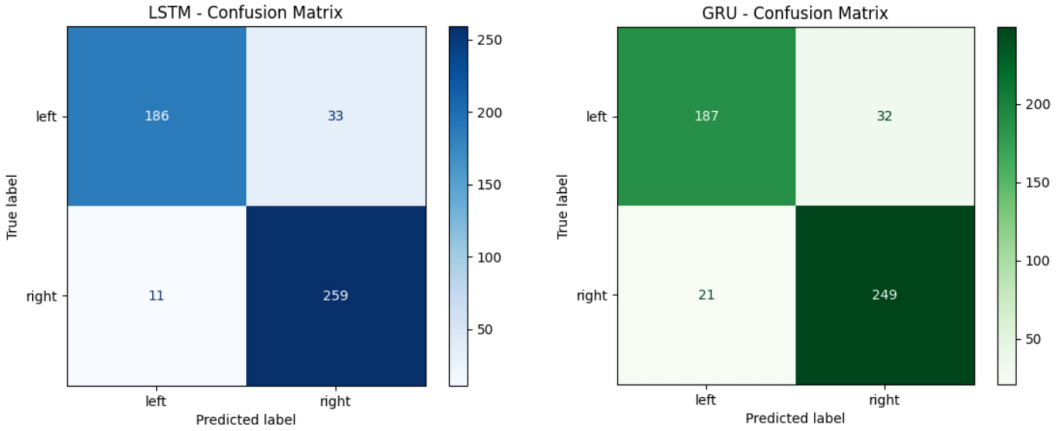
4.3 Sequential models

I also test an LSTM and a GRU model, trained on the de-biased corpus. The models use a vocabulary of 20,000 tokens, 128-dimensional embeddings, a hidden size of 128, dropout of 0.3, the Adam optimizer with learning rate 10^{-3}, and early stopping on validation performance.

Table 3: Comparison of the retained baseline and sequential models.

Model	Accuracy	F1
TF–IDF + Linear SVM	0.9900	0.9900
LSTM de-biased	0.9100	0.9217
GRU de-biased	0.8916	0.9038

The recurrent models learn a real political signal, but they remain clearly below the TF–IDF + SVM baseline. The LSTM performs better than the GRU, with an accuracy of 0.91 and an F1-score of about 0.92, compared with 0.8916 accuracy and about 0.90 F1 for the GRU. The confusion matrices confirm this gap: the LSTM correctly classifies 186 left-wing and 259 right-wing momentos, with 44 errors in total, while the GRU correctly classifies 187 left-wing and 249 right-wing momentos, with 53 errors. In both cases, more left-wing documents are predicted as right-wing than the reverse. This lower performance is likely due to the moderate size of the corpus and to the fact that the left–right distinction is strongly lexical. The recurrent models also show signs of overfitting, as training performance increases quickly while validation performance plateaus.



(a) LSTM confusion matrix

(b) GRU confusion matrix

Figure 3: Confusion matrices for the recurrent models.

Note: Panel (a) reports the confusion matrix for the LSTM model, while Panel (b) reports the confusion matrix for the GRU model. Both models capture the left–right distinction but make substantially more errors than the TF–IDF + SVM baseline.

4.4 Named-entity analysis

Finally, the NER analysis is used as an interpretive complement. The spaCy model detects 62,260 named entities in total, corresponding to an average of 24.68 entities per document. Locations are the most frequent entity type, followed by persons, organisations and miscellaneous entities.

Table 4: Distribution of named-entity types.

Entity type	Occurrences
LOC	23,382
PER	18,609
ORG	14,797
MISC	5,472

A difference also appears between the two political orientations. Right-wing momentos contain on average 29.21 entities per document, compared with 19.45 for left-wing momentos. The most frequent entities differ by class: left-wing texts often mention *Maastricht*, *Europe*, *PS*, *PCF*, Michel

Rocard and François Mitterrand, while right-wing texts more often mention *UDF*, *RPR*, *Front national*, *Union pour la France* and territorial references.

These results show that the left–right distinction is not only visible through generic vocabulary, but also through the actors, organisations and places mobilised in campaign discourse. The NER results must nevertheless be interpreted cautiously, since the model was not trained specifically on OCRed electoral momentos and some extracted entities remain noisy.

5 Discussion and limitations

The project shows that OCRed electoral momentos can be transformed into a supervised dataset and used for political classification. The strongest result is the performance of the TF–IDF + linear SVM baseline. This suggests that, in the 1993 legislative campaign, the left–right divide is strongly reflected in the vocabulary of electoral communication.

Nevertheless, several limitations remain. First, the quality of the OCRed texts is imperfect. OCR noise, layout artefacts, spelling variants and segmentation errors may affect both vectorisation and NER. Second, the labels are constructed from metadata rather than produced by expert manual annotation. The mapping is transparent and reproducible, but it remains an approximation of political orientation. Third, the focus on a single election year improves coherence but limits temporal generalisation. A classifier trained on 1993 may learn vocabulary specific to that political context.

Information leakage is another important limitation. Even if the de-biasing experiment is reassuring, some explicit political cues may remain in the texts. The high performance should therefore be interpreted as evidence of a strong political signal in this corpus, not as proof that all forms of political orientation could be predicted equally well without contextual cues.

The comparison with deep learning models also has to be interpreted carefully. LSTM and GRU models trained from scratch are useful benchmarks, but they are not necessarily the strongest neural approach for this task. Pre-trained French language models such as CamemBERT or FlauBERT could perform better, although they would require more computational resources and a different evaluation design.

Finally, the NER analysis is exploratory. It enriches the interpretation by showing which actors, organisations and places are mentioned by each political orientation, but the extracted entities contain noise. Future work could include entity post-processing, manual validation, or the integration of cleaned NER features into a hybrid classification model.

6 Conclusion

This report presents a complete NLP pipeline for classifying OCRed electoral momentos from the ARCHELEC corpus. The main contribution is not only the high classification score, but also the construction of a transparent labelled dataset from structured political metadata. After filtering the corpus on the first round of the 1993 French legislative elections, the final binary dataset contains 2,523 momentos labelled as *left* or *right*.

The TF–IDF + linear SVM baseline achieves a macro-F1 close to 0.99, showing that the left–right distinction is highly visible in the lexical structure of the corpus. The de-biasing experiment indicates that performance does not rely only on explicit party names or ideological markers. LSTM and GRU models capture a real signal but remain below the linear baseline, probably because the corpus is moderately sized and strongly structured by vocabulary. The NER analysis complements the classification results by highlighting the actors, organisations and places associated with each political orientation.

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